

MATH3881/5881: Statistical Machine Learning Theory

Week 3: Optimisation in Machine Learning: Foundations — Exercises and Solutions

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Exercises and Solutions

Exercise 3.0.1 (Ridge Regression Convergence)

Consider ridge regression: $C(\theta) = \frac{1}{n}\|y - X\theta\|^2 + \frac{\lambda}{2}\|\theta\|^2$ with $\lambda > 0$, where $X \in \mathbb{R}^{n \times d}$ has full column rank and $\lambda_{\max}, \lambda_{\min} > 0$ denote the largest and smallest eigenvalues of $X^\top X$.

- (a) Compute the Hessian $\nabla^2 C(\theta)$ and show C is L -smooth and μ -strongly convex with

$$L = \frac{2\lambda_{\max}}{n} + \lambda, \quad \mu = \frac{2\lambda_{\min}}{n} + \lambda.$$

- (b) Show that $\kappa(\lambda) = L/\mu \rightarrow \lambda_{\max}/\lambda_{\min}$ as $\lambda \rightarrow 0$ and $\kappa(\lambda) \rightarrow 1$ as $\lambda \rightarrow \infty$. Explain why regularisation speeds up gradient descent.
- (c) Let θ^* denote the unique minimiser of C (it exists and is unique because C is μ -strongly convex with $\mu > 0$). Using the strongly convex convergence theorem, find the number of iterations (with $\alpha = 1/L$) needed to guarantee $\|\theta^{(t)} - \theta^*\|^2 \leq \varepsilon \|\theta^{(0)} - \theta^*\|^2$. (*Hint: use $\log(1 - 1/\kappa) \geq -2/\kappa$ for $\kappa \geq 2$.*)
- (d) Suppose $\lambda_{\max}/\lambda_{\min} = 2500$. Without regularisation, estimate the iteration count for $\varepsilon = 10^{-6}$. How does adding $\lambda = 2\lambda_{\min}/n$ change the condition number and iteration count?

Solution.

- (a) Expanding: $C(\theta) = \frac{1}{n}\|y\|^2 - \frac{2}{n}y^\top X\theta + \frac{1}{n}\theta^\top X^\top X\theta + \frac{\lambda}{2}\|\theta\|^2$. The Hessian is

$$\nabla^2 C(\theta) = \frac{2}{n}X^\top X + \lambda I.$$

This is constant (independent of θ), so C is both L -smooth and μ -strongly convex with L and μ equal to the largest and smallest eigenvalues of $\nabla^2 C$. The eigenvalues of $X^\top X$ range from $\lambda_{\min}$ to $\lambda_{\max}$, so the eigenvalues of $\nabla^2 C$ range from $\frac{2\lambda_{\min}}{n} + \lambda$ to $\frac{2\lambda_{\max}}{n} + \lambda$, giving the stated L and μ .

$$(b) \quad \kappa(\lambda) = \frac{2\lambda_{\max}/n + \lambda}{2\lambda_{\min}/n + \lambda}.$$

$$\text{As } \lambda \rightarrow 0: \kappa \rightarrow \frac{2\lambda_{\max}/n}{2\lambda_{\min}/n} = \frac{\lambda_{\max}}{\lambda_{\min}}.$$

As $\lambda \rightarrow \infty$: numerator $\sim \lambda$ and denominator $\sim \lambda$, so $\kappa \rightarrow 1$.

The condition number measures how “elongated” the level sets of C are. Regularisation adds λI to the Hessian, raising all eigenvalues by λ and bringing them closer together in relative terms. This makes the objective more isotropic, so gradient descent does not oscillate along ill-conditioned directions and converges faster.

(c) The theorem gives $\|\theta^{(t)} - \theta^*\|^2 \leq (1 - \mu/L)^t \|\theta^{(0)} - \theta^*\|^2 = (1 - 1/\kappa)^t \|\theta^{(0)} - \theta^*\|^2$.

We need $(1 - 1/\kappa)^t \leq \varepsilon$. Using $1 - x \leq e^{-x}$ with $x = 1/\kappa$,

$$(1 - 1/\kappa)^t \leq e^{-t/\kappa},$$

so it suffices to take $t \geq \kappa \log(1/\varepsilon)$ (then $e^{-t/\kappa} \leq \varepsilon$). Conversely, the hint $\log(1 - 1/\kappa) \geq -2/\kappa$ for $\kappa \geq 2$ gives the matching *lower* bound on the required t : the exact threshold is at least $\frac{\kappa}{2} \log(1/\varepsilon)$. So $t = \Theta(\kappa \log(1/\varepsilon))$.

(d) Without regularisation: $\kappa = \lambda_{\max}/\lambda_{\min} = 2500$. For $\varepsilon = 10^{-6}$, the sufficient bound is

$$t \geq 2500 \times \log(10^6) = 2500 \times 6 \log 10 \approx 2500 \times 13.8 \approx 34,500 \text{ iterations.}$$

With $\lambda = 2\lambda_{\min}/n$:

$$L' = \frac{2\lambda_{\max}}{n} + \frac{2\lambda_{\min}}{n}, \quad \mu' = \frac{2\lambda_{\min}}{n} + \frac{2\lambda_{\min}}{n} = \frac{4\lambda_{\min}}{n},$$

$$\kappa' = \frac{\lambda_{\max} + \lambda_{\min}}{2\lambda_{\min}} = \frac{\lambda_{\max}/\lambda_{\min} + 1}{2} = \frac{2501}{2} \approx 1250.$$

The condition number halves (from 2500 to ≈ 1250), so the sufficient iteration count also halves: $t \geq 1250 \times 13.8 \approx 17,250$ iterations.

Exercise 3.0.2 (Optimal Step Size)

Recall that the one-step decrease bound, applied with general learning rate α , gives

$$C(\phi) \leq C(\theta) - \psi(\alpha) \|\nabla C(\theta)\|^2, \quad \psi(\alpha) := \alpha \left(1 - \frac{\alpha L}{2}\right).$$

The function $\psi(\alpha)$ is the *per-step decrease factor*: a larger value of $\psi(\alpha)$ means a larger guaranteed drop in C at each step.

- Show that $\psi(\alpha)$ is maximised over $\alpha > 0$ at $\alpha^* = 1/L$, giving maximum guaranteed decrease $\frac{1}{2L} \|\nabla C(\theta)\|^2$ per step.
- Show that $\psi(\alpha) > 0$ if and only if $\alpha \in (0, 2/L)$. Hence identify the full range of step sizes for which gradient descent is guaranteed to decrease C at every point where $\nabla C(\theta) \neq 0$.
- For a convex L -smooth function, how does the iteration count to achieve $C(\theta^{(t)}) - C(\theta^*) \leq \varepsilon$ scale if the step size is set to $\alpha = 1/(2L)$ instead of $\alpha = 1/L$? (*Hint*: redo the proof sketch with a general α .)

Solution.

- (a) Differentiating $\psi(\alpha) = \alpha - \frac{\alpha^2 L}{2}$ with respect to α :

$$\psi'(\alpha) = 1 - \alpha L.$$

Setting $\psi'(\alpha) = 0$ gives $\alpha^* = 1/L$. Since $\psi''(\alpha) = -L < 0$, this is a maximum. Substituting:

$$\psi(1/L) = \frac{1}{L} \left(1 - \frac{1}{2} \right) = \frac{1}{2L}.$$

Hence the maximum guaranteed decrease per step is $\frac{1}{2L} \|\nabla C(\theta)\|^2$.

- (b) For $\alpha > 0$, $\psi(\alpha) = \alpha(1 - \alpha L/2) > 0$ if and only if $1 - \alpha L/2 > 0$, i.e., $\alpha < 2/L$. Thus $\psi(\alpha) > 0 \iff \alpha \in (0, 2/L)$.

Gradient descent is guaranteed to strictly decrease C at every point where $\nabla C(\theta) \neq 0$ for any step size $\alpha \in (0, 2/L)$.

- (c) Retrace the proof for general $\alpha \leq 1/L$. From the descent lemma, $C(\theta^{(t+1)}) \leq C(\theta^{(t)}) - \alpha(1 - \alpha L/2) \|\nabla C(\theta^{(t)})\|^2$, so

$$C(\theta^{(t)}) - C(\theta^{(t+1)}) \geq \alpha(1 - \alpha L/2) \|\nabla C\|^2 \geq \frac{\alpha}{2} \|\nabla C\|^2$$

for $\alpha \leq 1/L$. This implies $\alpha^2 \|\nabla C\|^2 \leq 2\alpha(C(\theta^{(t)}) - C(\theta^{(t+1)})) = 2\alpha(\delta_t - \delta_{t+1})$, where $\delta_t = C(\theta^{(t)}) - C(\theta^*)$. Using the expansion of $r_{t+1} = \|\theta^{(t+1)} - \theta^*\|^2$,

$$r_{t+1} \leq r_t - 2\alpha\delta_t + \alpha^2 \|\nabla C\|^2 \leq r_t - 2\alpha\delta_{t+1},$$

so $\delta_{t+1} \leq \frac{r_t - r_{t+1}}{2\alpha}$. Summing from $s = 0$ to $t - 1$ and dropping $r_t \geq 0$:

$$\sum_{s=1}^t \delta_s \leq \frac{r_0}{2\alpha}.$$

Since δ_t is non-increasing, $t \delta_t \leq \sum_{s=1}^t \delta_s \leq r_0/(2\alpha)$, yielding

$$C(\theta^{(t)}) - C(\theta^*) \leq \frac{r_0}{2\alpha t}.$$

For $\alpha = 1/L$: the bound is $\frac{Lr_0}{2t}$, requiring $t \geq \frac{Lr_0}{2\varepsilon}$. For $\alpha = 1/(2L)$: the bound is $\frac{Lr_0}{t}$, requiring $t \geq \frac{Lr_0}{\varepsilon}$ — *twice as many iterations*.

Exercise 3.0.3 (Reverse-Mode Autodiff)

Let $C(\theta_1, \theta_2) = \theta_1 \cdot \exp(\theta_1 + \theta_2^2)$. Compute ∇C at $(\theta_1, \theta_2) = (1, 1)$ using reverse-mode autodiff with intermediate nodes $a_1 = \theta_2^2$, $a_2 = \theta_1 + a_1$, $a_3 = \exp(a_2)$, $C = \theta_1 \cdot a_3$.

- Introduce the intermediate variables and draw the computational graph.
- Perform the forward pass at $(1, 1)$.
- Apply the adjoint recursion to compute all adjoints and hence ∇C .
- Verify by direct differentiation.

Solution.

(a) The computational graph has leaf nodes θ_1 and θ_2 . The edges are: $\theta_2 \rightarrow a_1 = \theta_2^2$; $\theta_1, a_1 \rightarrow a_2 = \theta_1 + a_1$; $a_2 \rightarrow a_3 = e^{a_2}$; $\theta_1, a_3 \rightarrow C = \theta_1 \cdot a_3$. Note that θ_1 has two outgoing edges: to a_2 and directly to C .

(b) Forward pass at $(\theta_1, \theta_2) = (1, 1)$:

$$a_1 = 1, \quad a_2 = 1 + 1 = 2, \quad a_3 = e^2, \quad C = 1 \cdot e^2 = e^2.$$

(c) Adjoints (initialise $\zeta_C = 1$):

$$\begin{aligned} \zeta_{a_3} &= \frac{\partial C}{\partial a_3} = \theta_1 = 1, \\ \zeta_{a_2} &= \zeta_{a_3} \cdot \frac{\partial a_3}{\partial a_2} = 1 \cdot e^{a_2} = e^2, \\ \zeta_{a_1} &= \zeta_{a_2} \cdot \frac{\partial a_2}{\partial a_1} = e^2 \cdot 1 = e^2. \end{aligned}$$

For $\partial C / \partial \theta_2$: θ_2 only enters through a_1 , so

$$\frac{\partial C}{\partial \theta_2} = \zeta_{a_1} \cdot \frac{\partial a_1}{\partial \theta_2} = e^2 \cdot 2\theta_2 = 2e^2.$$

For $\partial C / \partial \theta_1$: θ_1 contributes via two paths — through a_2 and directly as a factor in $C = \theta_1 a_3$:

$$\frac{\partial C}{\partial \theta_1} = \underbrace{\zeta_{a_2} \cdot 1}_{\text{path via } a_2} + \underbrace{\zeta_C \cdot a_3}_{\text{direct path}} = e^2 + e^2 = 2e^2.$$

(d) Direct differentiation of $C = \theta_1 e^{\theta_1 + \theta_2^2}$:

$$\frac{\partial C}{\partial \theta_1} = e^{\theta_1 + \theta_2^2} + \theta_1 e^{\theta_1 + \theta_2^2} = (1 + \theta_1) e^{\theta_1 + \theta_2^2},$$

$$\frac{\partial C}{\partial \theta_2} = 2\theta_1 \theta_2 e^{\theta_1 + \theta_2^2}.$$

At $(1, 1)$: both equal $2e^2$, matching the reverse-mode result.

Exercise 3.0.4 (Concrete Backpropagation)

Consider a two-layer FNN ($L = 2$) with $N_0 = 2$, $N_1 = 2$, $N_2 = 1$. Hidden layer uses ReLU; output layer is linear. Loss: $C = \frac{1}{2}(a^{[2]} - y)^2$.

- (a) Write the full forward pass.
- (b) Show $\delta^{[2]} = a^{[2]} - y$.
- (c) Derive $\delta^{[1]}$ via the delta recursion.
- (d) With $z^{[1]} = (-1, 3)^\top$, give numerical values of $\dot{\sigma}_{\text{ReLU}}(z_j^{[1]})$ and simplify $\delta^{[1]}$.
- (e) State dimensions and give explicit outer-product formulas for $g_W^{[2]}$ and $g_W^{[1]}$.

Solution.

- (a) Full forward pass (with $a^{[0]} = x \in \mathbb{R}^2$):

$$\begin{aligned} z^{[1]} &= W^{[1]}x + b^{[1]} \in \mathbb{R}^2, \\ a^{[1]} &= \text{ReLU}(z^{[1]}) = \max(z^{[1]}, 0) \in \mathbb{R}^2, \\ z^{[2]} &= W^{[2]}a^{[1]} + b^{[2]} \in \mathbb{R}, \\ a^{[2]} &= z^{[2]} \in \mathbb{R} \quad (\text{linear output}), \\ C &= \frac{1}{2}(a^{[2]} - y)^2. \end{aligned}$$

- (b) The output-layer delta is $\delta^{[2]} = \partial C / \partial z^{[2]}$. By the chain rule:

$$\delta^{[2]} = \frac{\partial C}{\partial a^{[2]}} \cdot \frac{\partial a^{[2]}}{\partial z^{[2]}} = (a^{[2]} - y) \cdot 1 = a^{[2]} - y.$$

- (c) Applying the delta recursion with $\ell = 1$:

$$\delta^{[1]} = \text{Diag}(\dot{\sigma}_{\text{ReLU}}(z^{[1]})) W^{[2]\top} \delta^{[2]} \in \mathbb{R}^2,$$

where $\dot{\sigma}_{\text{ReLU}}(u) = \mathbf{1}[u > 0]$ is applied entry-wise.

- (d) With $z^{[1]} = (-1, 3)^\top$:

$$\dot{\sigma}_{\text{ReLU}}(z_1^{[1]}) = \dot{\sigma}_{\text{ReLU}}(-1) = 0, \quad \dot{\sigma}_{\text{ReLU}}(z_2^{[1]}) = \dot{\sigma}_{\text{ReLU}}(3) = 1.$$

Hence $\text{Diag}(\dot{\sigma}_{\text{ReLU}}(z^{[1]})) = \text{Diag}(0, 1)$, and writing $W^{[2]} = (w_1^{[2]}, w_2^{[2]})$:

$$\delta^{[1]} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} w_1^{[2]} \\ w_2^{[2]} \end{pmatrix} \delta^{[2]} = \begin{pmatrix} 0 \\ w_2^{[2]} \delta^{[2]} \end{pmatrix}.$$

The first hidden neuron is dead (its pre-activation is negative), so it contributes no gradient.

(e) From the weight-gradient formula $g_W^{[\ell]} = \delta^{[\ell]}(a^{[\ell-1]})^\top$:

$$g_W^{[2]} = \delta^{[2]}(a^{[1]})^\top \in \mathbb{R}^{1 \times 2} \quad (\text{outer product of scalars/vectors}),$$

$$g_W^{[1]} = \delta^{[1]}(a^{[0]})^\top = \delta^{[1]}x^\top \in \mathbb{R}^{2 \times 2}.$$

With $z^{[1]} = (-1, 3)^\top$, the first row of $g_W^{[1]}$ is all zeros (dead neuron).

Exercise 3.0.5 (Permutation Symmetry and Non-Convexity)

Let $P \in \mathbb{R}^{N_1 \times N_1}$ be a permutation matrix.

- (a) Show that replacing $(W^{[1]}, b^{[1]}, W^{[2]})$ by $(PW^{[1]}, Pb^{[1]}, W^{[2]}P^\top)$ produces the same output for every input x .
- (b) Explain why the $N_1!$ permuted parameter vectors are all global minimisers with the same loss.
- (c) With $N_1 = 2$, explain why the midpoint $\theta_{\text{mid}} = \frac{1}{2}(\theta^* + \tilde{\theta})$ typically has higher loss, and why this violates convexity.

Solution.

- (a) Under the permuted parameters, the hidden pre-activations become $PW^{[1]}x + Pb^{[1]} = P(W^{[1]}x + b^{[1]}) = Pz^{[1]}$. Since S (ReLU or any elementwise activation) commutes with permutations (it acts entry-wise):

$$\tilde{a}^{[1]} = S(Pz^{[1]}) = PS(z^{[1]}) = Pa^{[1]}.$$

The output then becomes:

$$\tilde{z}^{[2]} = W^{[2]}P^\top \cdot Pa^{[1]} + b^{[2]} = W^{[2]}(P^\top P)a^{[1]} + b^{[2]} = W^{[2]}a^{[1]} + b^{[2]} = z^{[2]},$$

using $P^\top P = I$. Hence the output is identical for every input x .

- (b) Since the network output is unchanged by any permutation of the hidden neurons, the loss $C(\theta)$ is the same for all $N_1!$ permuted parameter vectors. If θ^* is a global minimiser achieving the minimum loss value, then so are all permuted versions $\theta_P^* = (PW^{[1]}, Pb^{[1]}, W^{[2]}P^\top)$ — they are also global minimisers with the same loss.
- (c) Let $N_1 = 2$ and let $\tilde{\theta}$ be the permutation of θ^* that swaps the two hidden neurons ($\tilde{\theta} \neq \theta^*$ since the neurons are distinct). The midpoint $\theta_{\text{mid}} = \frac{1}{2}(\theta^* + \tilde{\theta})$ corresponds to a network in which both hidden neurons have identical (averaged) weights $\frac{1}{2}(w_1 + w_2)$ and biases $\frac{1}{2}(b_1 + b_2)$.

This “collapsed” network has effectively only one distinct neuron pattern (the two neurons are identical), which drastically limits its representational capacity. It cannot generally fit the training data as well as θ^* , so typically $C(\theta_{\text{mid}}) > C(\theta^*)$.

But $C(\theta^*) = C(\tilde{\theta})$, so:

$$C\left(\frac{\theta^* + \tilde{\theta}}{2}\right) > C(\theta^*) = \frac{1}{2}(C(\theta^*) + C(\tilde{\theta})),$$

which is the *reverse* of the convexity inequality. Hence C is not convex.

Exercise 3.0.6 (Vanishing Gradient Analysis)

Consider a uniform L -layer network with activation σ and weight spectral norms $\|W^{[\ell]}\|_2 = c$.

- (a) Show $\sigma'_{\text{Sig}}(u) \leq 1/4$ for all u , with equality at $u = 0$.
- (b) Derive the bound $\|\delta^{[1]}\|_2 \leq (c/4)^{L-1} \|\delta^{[L]}\|_2$ for sigmoid activations.
- (c) For a 20-layer sigmoid network with $c = 1$, evaluate $(1/4)^{19}$. What c prevents vanishing?
- (d) For ReLU with $c = 1$, state the Jacobian bound and identify the remaining failure mode.

Solution.

- (a) We have $\sigma'_{\text{Sig}}(u) = \sigma_{\text{Sig}}(u)(1 - \sigma_{\text{Sig}}(u))$. Let $s = \sigma_{\text{Sig}}(u) \in (0, 1)$. By the AM-GM inequality:

$$s(1 - s) \leq \left(\frac{s + (1 - s)}{2}\right)^2 = \frac{1}{4},$$

with equality when $s = 1 - s$, i.e. $s = 1/2$, which occurs at $u = 0$.

- (b) For each layer k , using submultiplicativity and the entry-wise bound $|\dot{\sigma}_{\text{Sig}}(u)| \leq 1/4$:

$$\|J^{[k]}\|_2 = \|\text{Diag}(\dot{\sigma}(z^{[k]})) W^{[k+1]^\top}\|_2 \leq \|\dot{\sigma}\|_\infty \cdot \|W^{[k+1]}\|_2 \leq \frac{c}{4}.$$

The Jacobian product formula gives $\delta^{[1]} = J^{[1]} J^{[2]} \dots J^{[L-1]} \delta^{[L]}$, so

$$\|\delta^{[1]}\|_2 \leq \prod_{k=1}^{L-1} \|J^{[k]}\|_2 \cdot \|\delta^{[L]}\|_2 \leq \left(\frac{c}{4}\right)^{L-1} \|\delta^{[L]}\|_2.$$

- (c) With $c = 1$, $L = 20$:

$$\left(\frac{1}{4}\right)^{19} = 4^{-19} \approx 3.64 \times 10^{-12}.$$

The gradient arriving at layer 1 is $\approx 10^{-12}$ times the output delta — effectively zero. Layer 1 receives no useful training signal; it cannot learn.

To prevent vanishing (keep the bound ≥ 1), we need $(c/4)^{19} \geq 1$, i.e. $c \geq 4$. But pushing weight norms this high to compensate for sigmoid saturation is not a viable training strategy; in practice the activation function is replaced (next part) rather than the weights inflated.

(d) For ReLU: $\dot{\sigma}_{\text{ReLU}}(u) \in \{0, 1\}$, so $\|\dot{\sigma}\|_{\infty} = 1$, giving

$$\|\delta^{[1]}\|_2 \leq (1 \cdot c)^{L-1} \|\delta^{[L]}\|_2 = 1^{19} \|\delta^{[L]}\|_2 = \|\delta^{[L]}\|_2.$$

This is a major improvement: gradients are bounded and do not vanish due to activation saturation. The remaining failure mode is **dead neurons**: if $z_j^{[\ell]} \leq 0$ for all training inputs throughout training, then $\dot{\sigma}_{\text{ReLU}}(z_j^{[\ell]}) = 0$ and no gradient flows through that neuron — it never recovers. This is partly addressed by careful initialisation (Xavier / He, Week 2).